

Do all from Slides (with Book for any explanation)

Exercise q's from notebook
↑ and orally.

LECTURE - 1

20/8/24
Tuesday.

⇒ CHAPTER 3: Vectors

• Page: 40 - 43 (till checkpoint 2)

LECTURE - 2

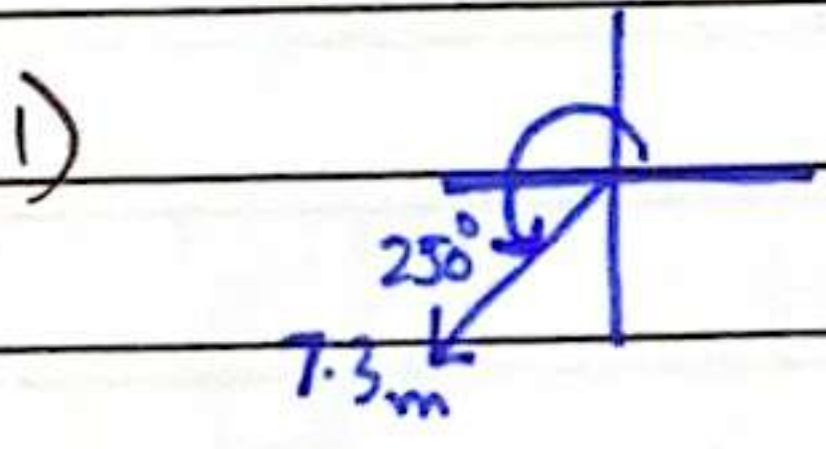
22/8/24
Thursday.

• Page: 44 - 55 (chapter complete)



* In head to tail diagram, the angle θ will be measured from tail of vector.

⇒ Exercise Questions: (onwards pg: 56)

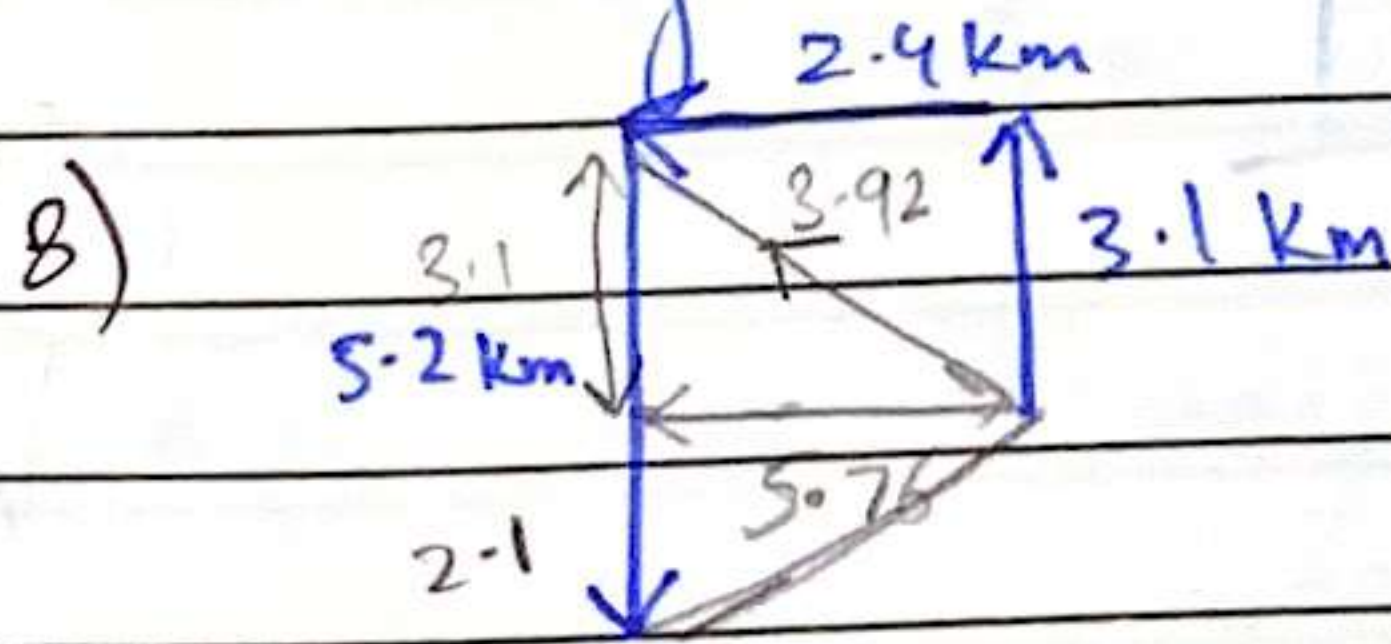


$$250 - 180 = 70^\circ$$

$$\begin{aligned} x\text{-comp} &= 7.3 \cos(70) = 3.31\text{m} \\ y\text{-comp} &= 7.3 \sin(70) = 6.50\text{m} \end{aligned}$$

6)

$$\begin{aligned} \text{horizontally} &= 12.5 \cos(20) = 11.89\text{m} \\ \text{vertically} &= 12.5 \sin(20) = 3.86\text{m} \end{aligned}$$



$$\begin{aligned} \sqrt{5.76^2 + 2.1^2} &= d \\ d &= 6.13\text{m} \text{ South west} \end{aligned}$$

Q9 (a) $(3m)i - (2m)j + (5m)k$

(b) $(4i - 3j + k) - (-i + j + 4k)$
 $= 5i - 4j - 3k$

(c) $\vec{c} = \vec{b} - \vec{a}$
 $= (-i + j + 4k) - (4i - 3j + k)$
 $= [-5i + 4j + 3k]$



13) Min distance = n comp + y comp.
 $= 3.4 \cos(35) + 3.4 \sin(35)$
 $= 14.68 \text{ km}$

15) (a) $x \cos(30) + 10 \cos(35) = 13.69 \text{ m}$

(b) $y \sin(30) + 10 \sin(35) = 13.07 \text{ m}$

$r = \sqrt{(3.69)^2 + (13.07)^2} = 13.58$

Angle $\rightarrow \tan^{-1}\left(\frac{13.07}{3.69}\right) = 82.5^\circ$

34) (a) $a \times b =$

i	j	k
3	5	0
2	4	6

 $\rightarrow$

k	35	24
0	24	0

ans: $2k$

(b) $a \cdot b = 6i + 20j$

(c) $a + b = (5i + 9j) \cdot (2i + 4j)$
 $= [10i + 36j]$

38) $3(7i - 8j) \cdot [(2(2i + 3j - 4k) \times (-3i + 4j + 2k))]$

$= (21i - 24j) \cdot [(4i + 6j - 8k) \times (3i + 4j + 2k)]$

$= (21i - 24j) \cdot$

i	j	k
4	6	-8
-3	4	2

i	j	k
6-8	-4	42
42	42	-3

$= (21i - 24j) \cdot (44i - 8j + 16k + 24j + 18k)$
 $= [924i - 384j]$

(4) $r = a + b + c + d$

$r_x = a_x + b_x + c_x + d_x \rightarrow -140 = 20 + b_x - 20 - 60$
 $b_x = -80$

$r_y = a_y + b_y + c_y + d_y \rightarrow 30 = 60 - 70 + c_y - 70$
 $c_y = 110 \text{ m}$

$|r| = \sqrt{(r_x^2 + r_y^2)}$
 $|r| = \sqrt{(-140)^2 + (30)^2} = 143.17 \text{ m}$

direction: $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\theta = \tan^{-1}\left(\frac{30}{-140}\right) = +12^\circ \rightarrow$ Basic angle

from the x-axis $\rightarrow 180 - 12$ (since 2nd Quad)
 direction $= 168^\circ$

$v_{0x} = v_0 \cos \theta_0$ (initial x velocity) $v_x = v_0 \cos \theta_0$ (final x velocity)
 $v_{0y} = v_0 \sin \theta_0$ (initial y velocity) $v_y \Rightarrow v_y = v_{iy} + at$
 $v_y = (v_0 \sin \theta_0) - (9.8)t$ (final y velocity)
 M T W T F S S
 Day: ☆☆☆☆☆☆
 Imp. (whenever xly component of velocity)

M T W T F S S
 Day: ☆☆☆☆☆☆
 Date: □□□□

Do relative chapter from Email
 slides & Register (open for reference)
 LECTURE - 3: Tuesday 27/8/24.

CHAPTER 4: Position & Displacement.

Page: 62-72 (till vertical comp. of Projectile)

LECTURE - 4: Thursday 29/8/24.

Horizontal Motion:

① $v_f = v_i + at$ $\rightarrow a=0$ (in horizontal)
 $v_i = v_0, v_f = v_x$
 $v_x = v_{0x} + 0$
 $v_x = v_0 \cos \theta_0$

② $S = v_i t + \frac{1}{2} at^2$ $a=0$

$x - x_0 = S \rightarrow$ for horizontal
 $x - x_0 = v_0 \cos \theta_0 t + 0 \rightarrow v_x = v_{0x}$
 $v_x = v_0 \cos \theta_0$

③ $2as = (v_f)^2 - (v_i)^2 + a=0$
 $0 = (v_x)^2 - (v_0 \cos \theta_0)^2$

$(v_x)^2 = (v_0 \cos \theta_0)^2$
 $v_x = v_{0x} = \text{constant} \rightarrow v_x = v_0 \cos \theta_0$

⇒ Eq. of Path Parabolic. $x_0 = y_0 = \text{initial values}$

From $s = ut + \frac{1}{2} at^2$ we break it into x & y components
 $x - x_0 = v_0 \cos \theta_0 t \rightarrow ①$
 $y - y_0 = v_0 \sin \theta_0 t - \frac{1}{2} gt^2 \rightarrow ②$

$t = \frac{x - x_0}{v_0 \cos \theta_0}$

* Put value of 't' in eq ②.

$y - y_0 = v_0 \sin \theta_0 \left(\frac{x - x_0}{v_0 \cos \theta_0} \right) - \frac{1}{2} g \left(\frac{x - x_0}{v_0 \cos \theta_0} \right)^2$

* let $x_0 = y_0 = 0$ (in start)

$y - 0 = \tan \theta_0 x - \frac{1}{2} g \left(\frac{x}{v_0 \cos \theta_0} \right)^2$

$y = ax + bx^2 \rightarrow$ general of path (parabola)
 $a = \tan \theta_0$
 $b = -\frac{g}{2(v_0 \cos \theta_0)^2}$

• $S = ut + \frac{1}{2} at^2$

- Horizontal $\rightarrow S_x = ut + \frac{1}{2} (a) t^2$

- Vertical $\rightarrow S_y = u_y t + \frac{1}{2} g t^2$

$S_y = (v_0 \sin \theta_0) t - \frac{1}{2} g t^2$
 core idea: horizontal displacement kay hoga & vertical kay liye "uy"

⇒ Range of Path: (complete)

$n - n_0 = v_0 \cos \theta_0 t$ — $R = n - n_0$

→ ①

* vertical distance wld be zero.

$0 = v_0 \sin \theta_0 t - 1/2 g t^2$ → ②

* value of 't' from ①

$t = \frac{R}{v_0 \cos \theta_0}$

* Put 't' in eq ②.

$0 = v_0 \sin \theta_0 \left(\frac{R}{v_0 \cos \theta_0} \right) - \frac{1}{2} g \left(\frac{R}{v_0 \cos \theta_0} \right)^2$

$\frac{1}{2} g \left(\frac{R}{v_0 \cos \theta_0} \right)^2 = v_0 \sin \theta_0 \left(\frac{R}{v_0 \cos \theta_0} \right)$

$\frac{1}{2} g \left(\frac{R}{v_0 \cos \theta_0} \right) = v_0 \sin \theta_0$

$R = \frac{2 \sin \theta_0 \cos \theta_0 (v_0)^2}{g}$

$R = \frac{(v_0)^2 \sin 2\theta_0}{g}$

⇒ Height of Path (complete)

From "Gas = $v^2 - v_x^2$ "

$H = y - y_0$

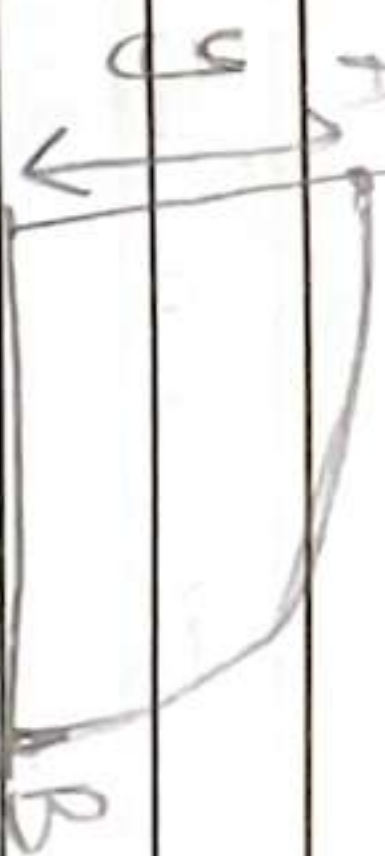
$-2g(y - y_0) = v_y^2 - (v_0 \sin \theta_0)^2$

$-2gH = 0 - (v_0 \sin \theta_0)^2$

$+ 2gH = (v_0 \sin \theta_0)^2$

$H = \frac{(v_0 \sin \theta_0)^2}{2g}$

⇒ Time of flight



At A, $v_y = 0$ m/s. (Therefore $v_x = 0$ m/s)

$S = v_x t + \frac{1}{2} g t^2$

* "S" since object moving back to position

$2y = g t^2$

$\sqrt{\frac{2y}{g}} = t$

CHAP: 4 completed. (Do 4.5 from Notes/Slides)

(4.6 & 4.7 Not included in Syllabus)

Not in Mid 2.2 or Finals

Marked ques. LECTURE-4 Tuesday 3/9/24.
on page 84.

Q7: $v_{avg} = \frac{dx}{dt}i + \frac{dy}{dt}j + \frac{dz}{dt}k$
 $= \frac{(-2-5)}{10}i + \frac{(8-(t^2))}{10}j + \frac{(-2-2)}{10}k$
 $= -0.7i + 1.4j - 0.4k$

Q11: (a) $t = 2$
 $\vec{v} = 6(2^2)i - 5(2^2)j + (6-7(2^2))k$
 $= 24i - 20j - 10k$

(b) $\vec{v} = (6t^2 - 5)i + (-28t^3)j$
 $\vec{r} = 19i - 22j$

(d) line of tangent so velocity $\left(\frac{dy}{dx}\right)$
 $\theta = \tan^{-1}\left(\frac{-22}{19}\right) = 85.2^\circ$

* Since 4th quad. $360 - 85.2 = 274.8^\circ$
 ans: $[274.8^\circ]$ or $[-85.2^\circ]$

Q25: $R = \frac{u^2 \sin 2\theta}{g}$
 $u = ?$

$\frac{77 \times 9.81}{\sin 2(17)} = u$
 $u = 43.1 \text{ m/s}$

Q28: (a) $s = ut + \frac{1}{2}at^2$
 After $t = 5.5$, s becomes 'h'
 $* u_y = u \sin \theta$

$h = \frac{(42 \sin 60)(5.5) + \frac{1}{2}(9.81)(5.5)^2}{2}$

(b) $\vec{v} = \sqrt{(u_y)^2 + (u_x)^2}$
 $u_x = 42 \cos 60 = 21$ → remains same for entire journey
 $u_y = 42 \sin 60 = 36.1$
 $v = 47.4 \text{ m/s}$

$u_y = u \sin \theta + at$
 $u_y = 42 \sin 60 + (-9.81)(5.5)$
 $u_y = -17.58 \text{ m/s}$

$\vec{v} = \sqrt{(21)^2 + (-17.58)^2}$
 $\vec{v} = 27.4 \text{ m/s}$

(c) $H = \frac{u^2 \sin^2 \theta}{g}$
 $H = \frac{(42)^2 \sin^2(60)}{9.81} = 67.4 \text{ m}$

$H = \frac{(42)^2 \sin^2(60)}{2(9.81)}$

Q33: a) $S_y = u_y t + \frac{1}{2}at^2$
 $0 - 730 = (u \sin \theta)t + \frac{1}{2}at^2$
 $-730 = u \sin(37)(5) + \frac{1}{2}(9.81)(5)^2$
 $u = 202 \text{ m/s}$
 from $u \sin \theta$

$-730 = u \sin(37)(5) + \frac{1}{2}(9.81)(5)^2$
 $u = 202 \text{ m/s}$

(b) * Not the range formula since this is just a part of projectile and is not even half projectile (we don't know).

$S_x = u_x t$
 $S_x = 202 \cos(37)(5)$
 $S_x = 807 \text{ m/s}$

(c) $u_x = u \cos \theta$

$u_x = u_x$ (since horizontal comp. doesn't change).

$$u_x = u \cos \theta = 202 \cos 37^\circ$$

$$u_x = 161 \text{ m/s}$$

$$v = u + at$$

$$v_y = u_y + (-9.81)(t) \rightarrow \text{negative with angle is very imp.}$$

$$v_y = 202 \sin(37^\circ) - (9.81)(5)$$

$$v_y = -171 \text{ m/s}$$

Thursday

5/9/24

LECTURE: 5

⇒ CHAPTER 5: Force & Motion - 1

* Do it completely from slides & sample problems from Book. (5.01, 5.02, 5.03, 5.04, 5.06)

Thursday
10/9/24

LECTURE: 6

⇒ Exercise: CHAPTER 5.

Q5: (a) Since we need acceleration in components so;

$$F_1 = (32 \cos 30^\circ) i + (32 \sin 30^\circ) j$$

$$F_1 = 27.7 i + 16 j$$

$$F_2 = (55 \cos 0^\circ) i + (55 \sin 0^\circ) j$$

$$F_2 = 55 i$$

$$F_3 = (41 \cos (-60^\circ)) i + (41 \sin (-60^\circ)) j$$

$$F_3 = (20.5) i + (-35.5) j$$

$$* a = F/m$$

$$a = (27.7 + 55 + 20.5) i + (16 + 0 - 35.5) j$$

$$* \text{Add } i \text{ and } j \text{ with } i \text{ and } j.$$

$$a = (27.7 + 55 + 20.5) i + (16 + 0 - 35.5) j$$

$$a = (0.86 i) + (-0.16 j) \text{ m/s}^2$$

* For q/s like these, use formula $\rightarrow F_1 + F_2 + F_3 = ma$.

* If acceleration is asked in unit vector notation, then we'll use i, j, k form of F_{net} . If just acceleration is asked, then we'll use the magnitude of F_{net} .

Thurs day
12/9/24

LECTURE: 7

From Notes and Man's Slides.

⇒ CHAP: 15 Oscillatory Motion

ONLY: 15.4, 15.1
ONLY: 15.4, 15.1
ONLY: 15.4, 15.1
ONLY: 15.4, 15.1

Motor

Book and Solution in Man's

PDF saved in 'Phone's

Files

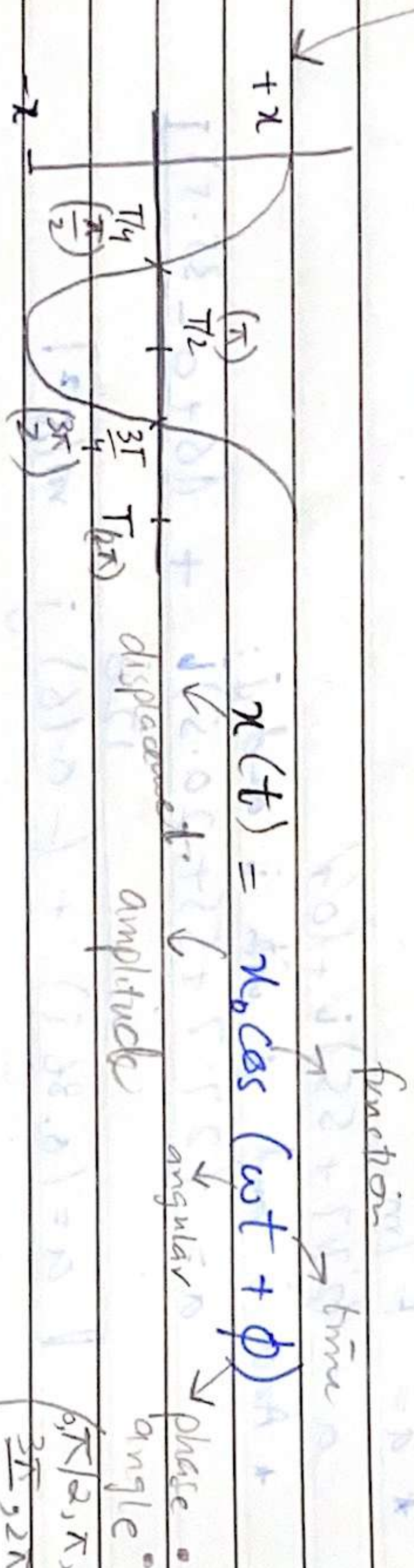
Oscillatory: Repeat after some time

Non-Oscillatory

SHM

Non SHM

- $a = -\omega^2 x$
- Damping of Pendulum



★ If func. is 'cos'; particle starting from 'extreme position'

if func. is 'sin'; particle starting from 'mean position'

$(\omega t + \phi)$ = Phase of motion

⇒ Derivation of $\omega = 2\pi f$

$$x(t) = x_0 \cos(\omega t + \phi)$$

At $t=0$, $\phi = 0^\circ$

Now $t=T$, $\phi = 2\pi$

$$\omega T = 2\pi$$

$$\omega(t+T) = \omega t + 2\pi$$

$$\omega t + \omega T = \omega t + 2\pi$$

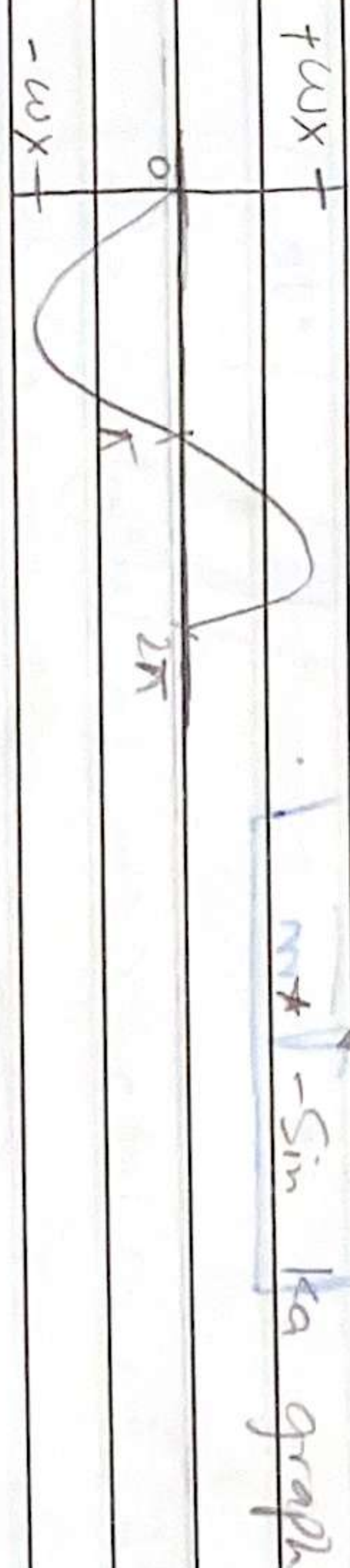
$$\omega T = 2\pi$$

$$\omega = \frac{2\pi}{T} = 2\pi f$$

⇒ velocity from displacement:

Take derivative of $x(t) = x_0 \cos(\omega t + \phi)$

$$v = -\omega x_0 \sin(\omega t + \phi)$$



⇒ velocity to acceleration:

Derivative of $v = -\omega x_0 \sin(\omega t + \phi)$

$$acc. = -\omega^2 x_0 \cos(\omega t + \phi)$$

★ Now graph will be of $-\cos$.

★ Whenever func. is changed from $\sin \rightarrow \cos$, $\cos \rightarrow \sin$
or we go from displacement $\rightleftharpoons$ velocity $\rightleftharpoons$ acceleration;
there will be phase diff (ϕ) of $\frac{\pi}{2}$ in each switch.

$\Rightarrow$ Force law: to find another value of ω .

$F = -kx \rightarrow \text{① } (T + \frac{1}{\omega})$

~~$F = -kx \rightarrow T + \frac{1}{\omega}$~~

$F = ma \rightarrow \text{② } T + \frac{1}{\omega}$

$a = -\omega^2 x \rightarrow \text{③ } T + \frac{1}{\omega}$

★ using ①, ②, ③

$F = m(-\omega^2 x)$

$(-k)x = m\omega^2 x$

$k = m\omega^2$

$\omega = \sqrt{\frac{k}{m}}$

$\omega = \sqrt{\frac{k}{m}}$

$\omega = \sqrt{\frac{k}{m}}$

$\omega = \sqrt{\frac{k}{m}}$

Thursday
19/9/24

LECTURE-8

$\rightarrow$ Torional Pendulum:

$F = -kx$

$F = ma$

① $\leftarrow \tau = -K\theta$

② $\leftarrow \tau = I\alpha$

using ① & ②

$I\alpha = -K\theta$

★ Since $a = \frac{d^2x}{dt^2}$ so ; $\alpha = \frac{d^2\theta}{dt^2}$

$\rightarrow I \left(\frac{d^2\theta}{dt^2} \right) = -K\theta \rightarrow$ Differential eq. of "torional Pendulum"

~~$I \left(\frac{d^2\theta}{dt^2} \right) + K\theta = 0$~~

$\frac{d^2\theta}{dt^2} + \frac{K}{I}\theta = 0$

$\frac{d^2x}{dt^2} + \omega^2 x = 0 \rightarrow$ Differential eq. of "SHM"

★ Comparing both differential eq.

$\omega^2 = \frac{K}{I}$

$T = 2\pi \sqrt{\frac{I}{K}}$

→ Simple Pendulum:

restoring torque

$\tau = r(LF)$ perpendicular to weight.

length of pendulum

$$\tau = -Lmg \sin \theta$$

* Since $\tau = I\alpha$

$$I\alpha = -Lmg \sin \theta$$

$$I\ddot{\theta} + Lmg \sin \theta = 0$$

$$\frac{d^2 \theta}{dt^2} + \frac{Lmg}{I} \sin \theta = 0 \rightarrow \text{differential eq. of simple pendulum}$$

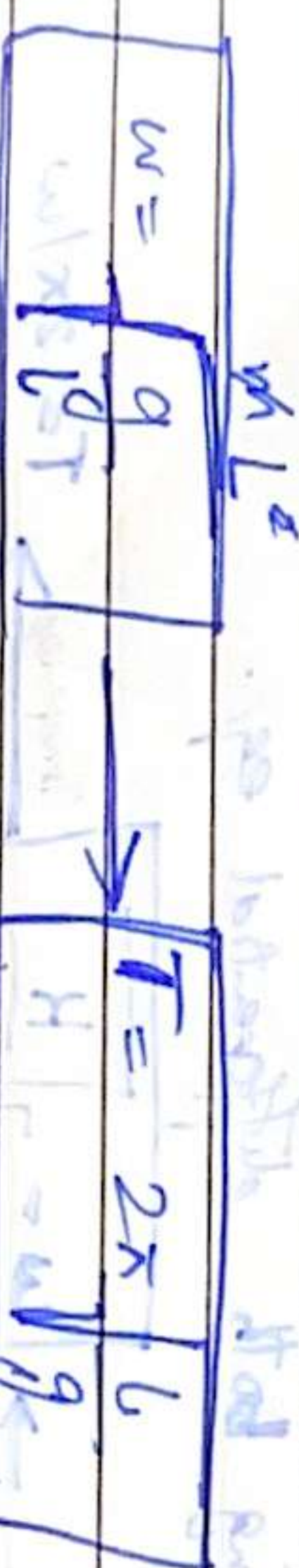
But since we're taking 'sin θ ' as 'very less displacement'.

* Comparing this differential with differential of SHM: $\frac{d^2 x}{dt^2} + \omega^2 x = 0$

Therefore; $\omega^2 = \frac{Lmg}{I}$

* Since $I = mL^2$

$$\omega = \sqrt{\frac{Lmg}{mL^2}} = \sqrt{\frac{g}{L}}$$



$$I = mL^2$$

CHAPTER 16: WAVES

LECTURE - 10 / Tuesday 1/10/24



→ Displacement Function:

$$y(x, t) = y_m \sin(kx - \omega t + \phi)$$

Amplitude wave number

Q: Find wave number (k):

* Suppose $t = 0$ and $\phi = 0$.

$$y(x, 0) = y_m \sin(kx)$$

* Moving λ distance in wave while keep $t = 0$ & $x = x + \lambda$

$$y(x + \lambda, 0) = y_m \sin(k(x + \lambda))$$

* Since λ distance is covered in $\sin(2\pi)$; therefore; $y(x + \lambda, 0) = y_m \sin(kx + 2\pi)$

$$k\lambda = 2\pi$$

$$k = \frac{2\pi}{\lambda}$$

Q: Find 'omega' (angular freq.):

* Suppose $x = 0$ and $\phi = 0$.

$$y(0, t) = y_m \sin(\omega t)$$

* Particle will come to 'y' time after it covered distance

in a timeperiod of wave i.e. 'T'. Therefore;

$$y(0, (t + T)) = y_m \sin(\omega(t + T))$$

★ Since one time period 'T' is covered when ~~the~~ a cycle is completed (i.e.: 2π) so;

$$\omega T = 2\pi$$

$$\omega = \frac{2\pi}{T}$$

★ The displacement func. remains constant even though we change the value of 'ω' and 'k'.

This is because ~~phase~~ $kx - \omega t = \text{constant}$

→ Taking its time derivative for "velocity".

$$k \left(\frac{dy}{dt} \right) - \omega = 0$$

★ $dy/dt = \text{velocity}$

$$kv - \omega = 0$$

$$v = \frac{\omega}{k}$$

value of $\leftarrow \frac{2\pi}{\lambda}$ from previous page

$$v = \frac{\omega}{k} = \frac{2\pi/T}{2\pi/\lambda} = \frac{\omega \lambda}{2\pi} = \frac{2\pi f \lambda}{2\pi} = f \lambda$$

displacement

(always +ve)

Amplitude

Oscillatory term

$$y = 2y_m \cos\left(\frac{1}{2}(\Phi)\right) \cdot \sin\left(kx - \omega t - \frac{\Phi}{2}\right)$$

$$y = \frac{2y_m \cos\left(\frac{\Phi}{2}\right) \sin\left(\frac{1}{2}(2kx - 2\omega t + \Phi)\right)}{2}$$

$$\cos\left(\frac{1}{2}(kx - \omega t - \frac{\Phi}{2})\right) = \cos\left(\frac{1}{2}(kx - \omega t + \Phi)\right)$$

$$y = 2y_m \left[\sin\left(\frac{1}{2}(kx - \omega t + kx - \omega t + \Phi)\right) \cdot \cos\left(\frac{1}{2}(kx - \omega t + \Phi)\right) \right]$$

$$\star \sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$$

$$y = y_m \left[\sin(kx - \omega t) + \sin(kx - \omega t + \Phi) \right]$$

$$y = y_m \sin(kx - \omega t) + y_m \sin(kx - \omega t + \Phi)$$

⇒ Interference: Adding displacement func. of 2 ↑ waves

LECTURE-11 Thursday 3/10/24

→ Constructive Interference:

Two waves overlapping.

* $\phi = 0$ (phase difference = 0)

* After putting it in previous eq, the displacement will be:

$$y = y_m \cos \left(\frac{2\pi y}{2y_m} \right)$$

Whenever $\phi = n\lambda$ (i.e. $0, 2\pi, 4\pi, 6\pi, 8\pi, \dots$): $y = 2y_m$.

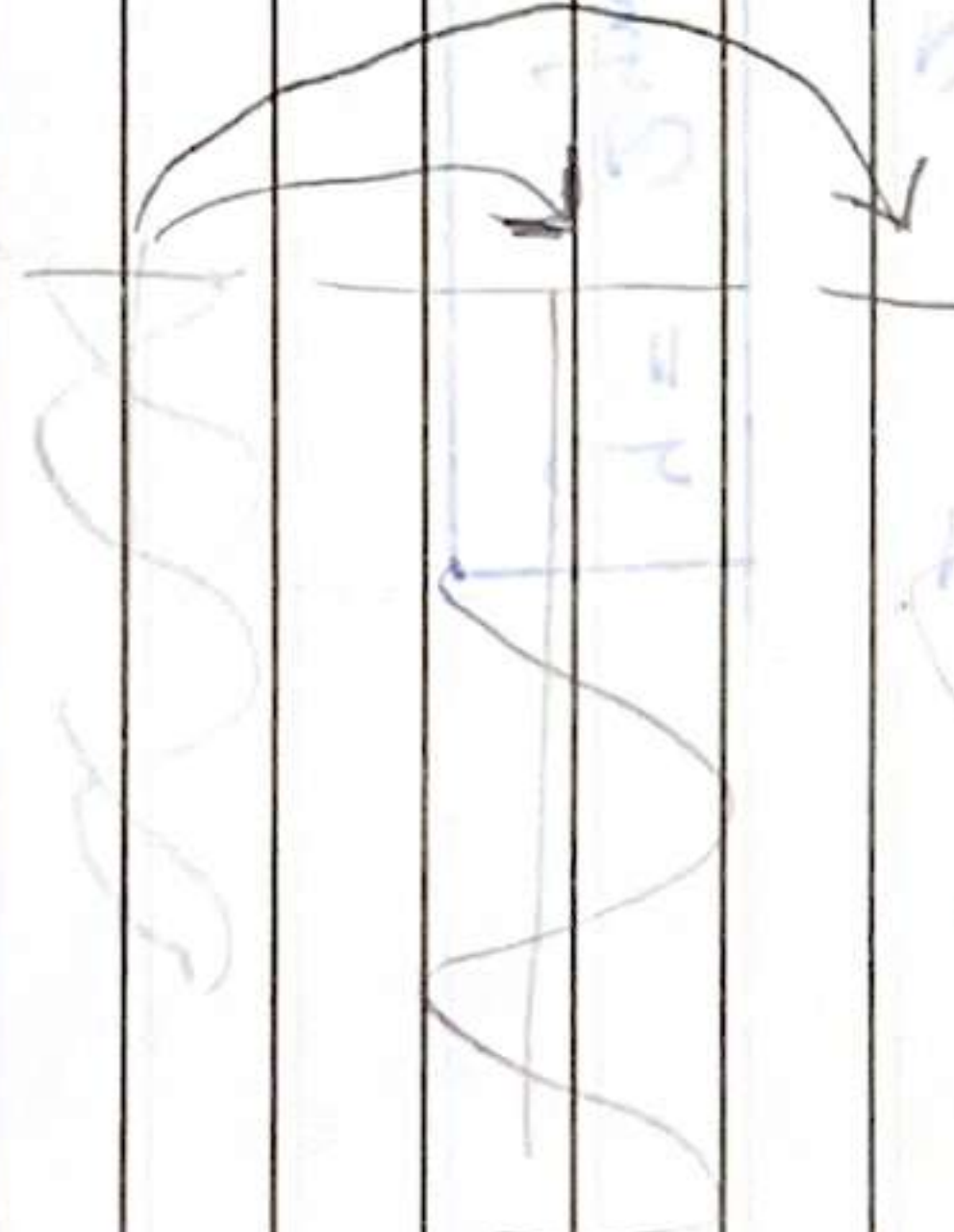
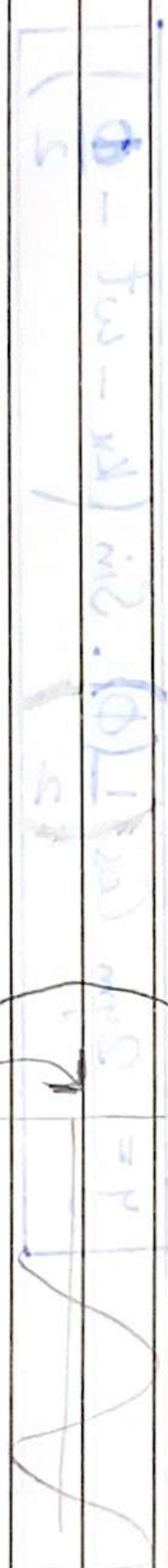
→ Destructive Interference:

* putting $\phi = \pi$ in general eq.

→ whenever $\phi = (n+1)\lambda$ (i.e. $\lambda, 3\lambda, 5\lambda, 7\lambda, \dots$): $y = 0$

$$y = y_m \cos \left(\frac{\pi}{2} \right) \sin \left(kx - \omega t - \frac{\pi}{2} \right)$$

$y = 0$ (wave will vanish).



Entire chapter, Sample problems:

1, 2, 3, 4, 5. Chapter problems:

34, 5, 6, 13, 15, 24, 25, 26, 27, 28, 41, 42.

⇒ Chapter 24: Coulomb's law.

LECTURE - 12

Tuesday
8/10/24

$$F \propto \frac{1}{r^2}$$

$$F = k \frac{|q_1| |q_2|}{r^2}$$

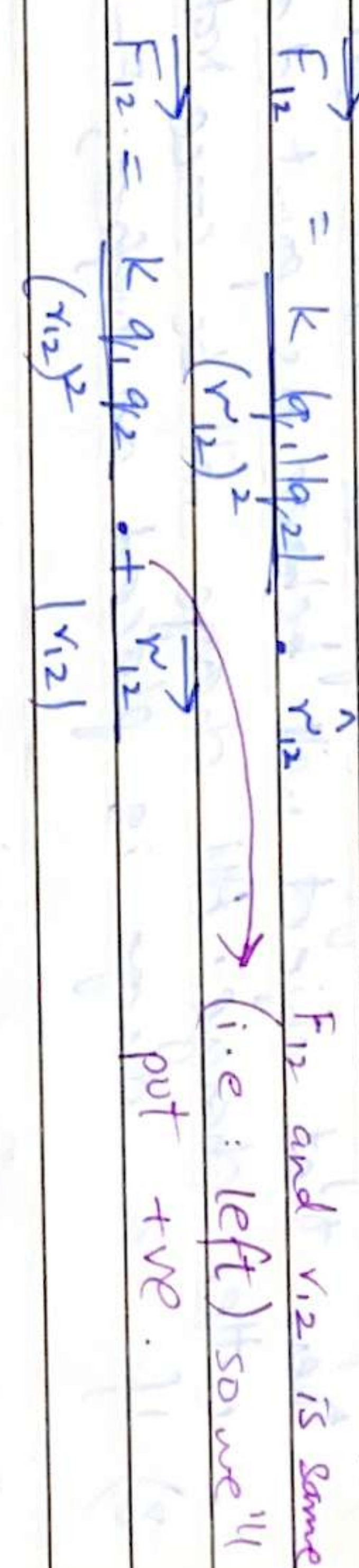
$$\text{vector } \vec{F} = k \frac{|q_1| |q_2|}{r^2} \cdot \hat{r}$$

→ this thing isn't periodically used anywhere.

* F_{12} force on charge 1 wrt charge 2.

* r_{12} distance of charge 1 wrt charge 2 (means we have to start measuring from 2 and put arrow head towards 1).

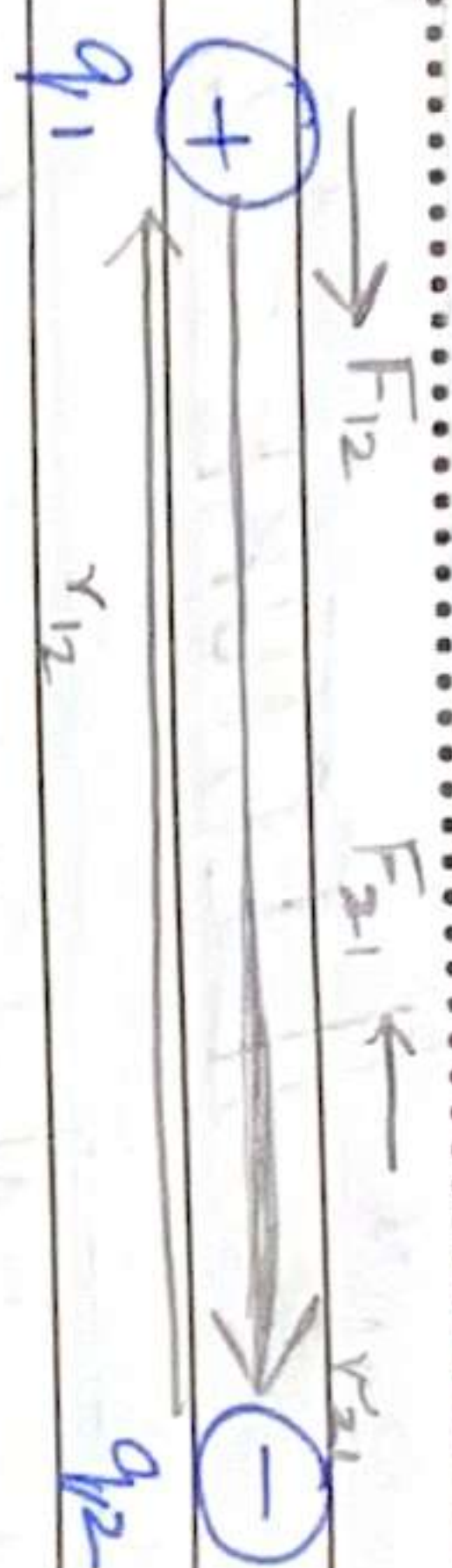
Case #1: (like charges)



$$\vec{F}_{12} = k \frac{|q_1| |q_2|}{r_{12}^2} \cdot \frac{\vec{r}_{12}}{r_{12}}$$

→ solved similarly.

Case #2



$$F_{12} = \frac{k q_1 q_2}{(r_{12})^2}$$

since direction of F_{12} and r_{12} is opposite, so we'll put -ve with r_{12}

→ Unit of k (Coulomb's constant) $9 \times 10^9 \text{ Nm}^2/\text{C}^2$

$$F = \frac{k q_1 q_2}{r^2} \quad N = \frac{k C^2}{m^2} \quad \boxed{k = \frac{m^2 \cdot N}{C^2}}$$

→ Shell's Theorem:

- 1) whenever an object is charged, the charge is spread uniformly on the surface of object. Therefore, if a charge is placed inside that object, that charge experience zero force.
- 2) If any charge is placed outside that object then that object will behave as point charge for outer charge. All charge will be concentrated at center.
- 3) If a charge is placed on surface; $F = \text{Maximum}$.

→ Charge Quantization:

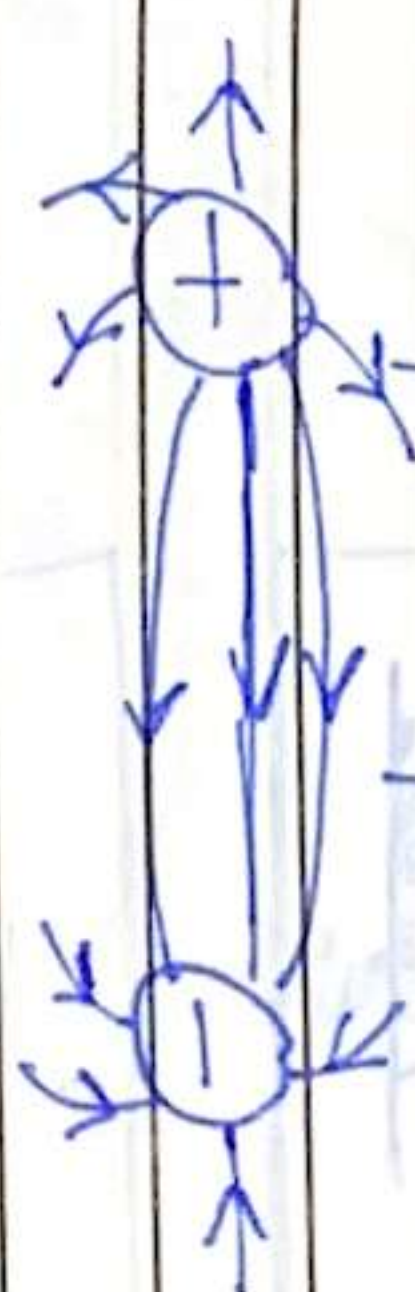
$$q = n e \rightarrow \text{elementary charge}$$

$\hookrightarrow \pm 1, \pm 2, \pm 3 \dots$

→ 22.1, 22.2, 22.3, 22.4
→ Problems: 3, 5, 7, 11, 20

⇒ CHAPTER # 22: "Electric Field"

→ Dipole: When we have 1 opposite charges of same magnitude and ~~same~~ specific constant distance, then the two fields of both charges will create dipole.

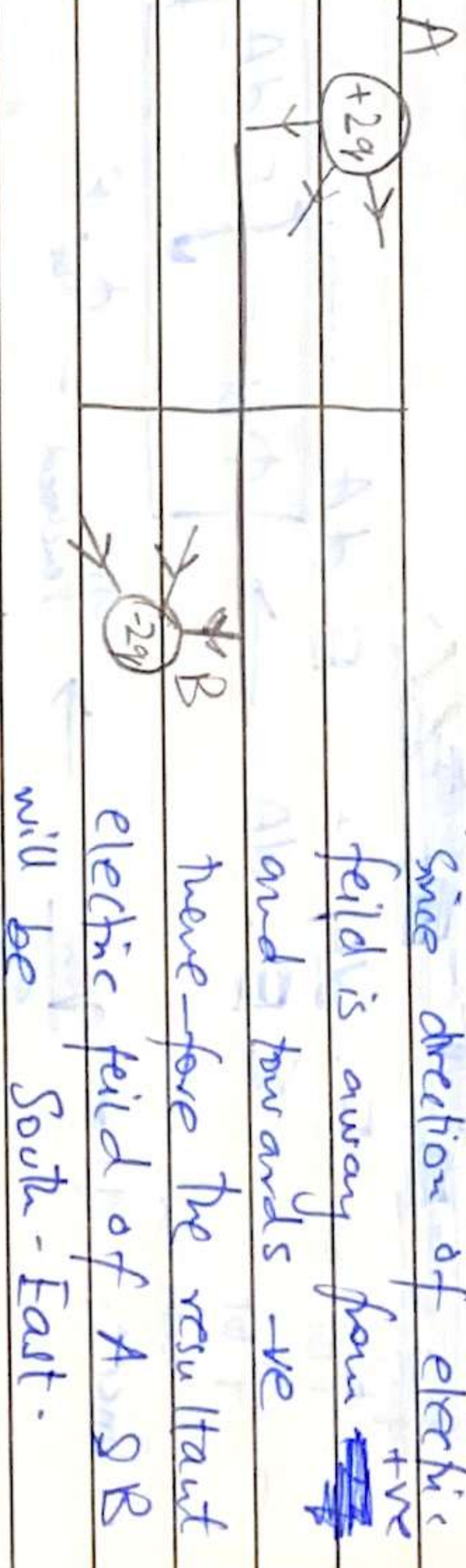


★ Strength of dipole is max. in blue.

→ Electric field of Dipole: (from Notes)

★ Dipole moment is always from +ve to -ve.

Diagram of Net electric Field (Sample problem: 22.01)



→ Derivation of Electric field due to dipole on page after chapter-23 (after 3 pages).

Tuesday (online)

15/10/24

$$E = \frac{q}{2\epsilon_0} \rightarrow \text{Planner symmetry, Non conducting}$$

$E = \frac{26}{20} \rightarrow$ Planner symmetry:
2 conducting plates.
Date ☐ ☐ ☐

77105 10

76105E-10

$q_{\text{enc}} = \oint E \cdot dA \rightarrow$ Gauss's law.

100

Since Electric field is constant so we'll put it

out of $\mathbb{R}^n$ then

2

Area of sphere = $4\pi r^2$

100

zelechno feld

Answers to 11 problems

$$P = EA \cos \theta \rightarrow \text{Max. when } \theta = 0^\circ.$$

→ M_{in} when $Q = q_c$

① $\Rightarrow$ Electric field of charged conductor: $\perp$ to surface

$$q_{\text{enc}} = \oint E \cdot dA$$

1. 1.1

Electric field uniform so treated as constant

$$g_{\text{Herc}} = 20 \cdot \frac{f_{\text{H}}}{\lambda} \rightarrow \text{area of conductor}$$

$$g_{\text{pure}} = \epsilon_0 \cdot A \cdot E$$

surface charge density (σ) = q/L^2

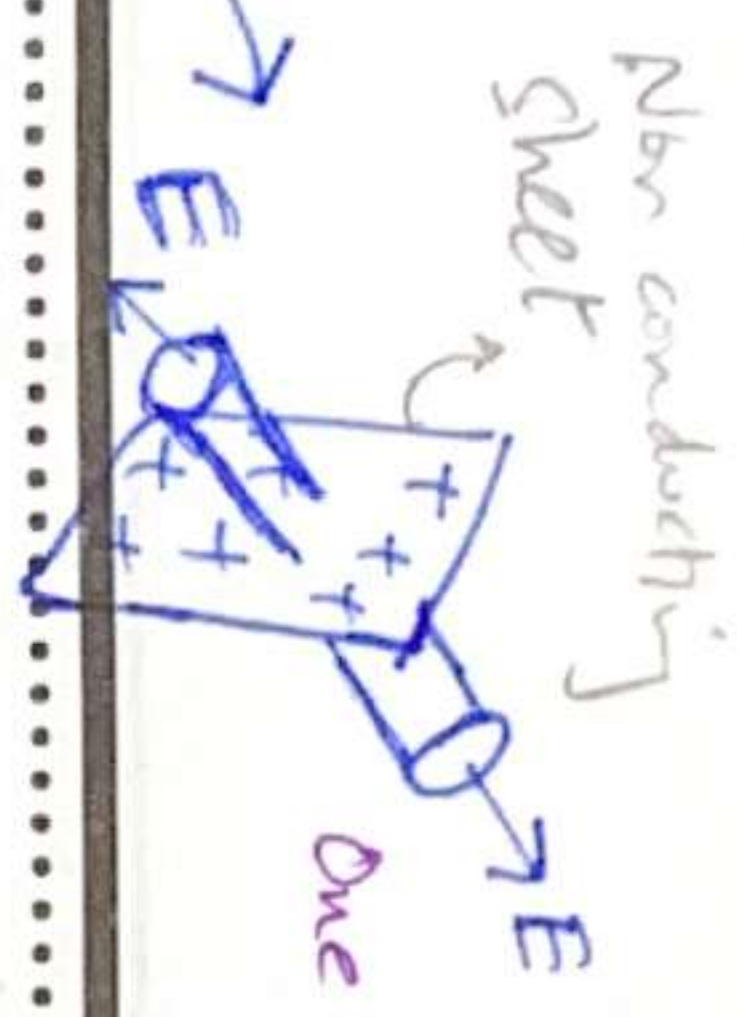
$$x_{\text{gen}} = 0.1$$

6. 11. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300. 301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400. 401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500. 501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600. 601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700. 701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800. 801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 840. 841. 842. 843. 844. 845. 846. 847. 848. 849. 850. 851. 852. 85

[illegible]

10

10



② Non - Conducting Plane: have charge spread on each side/face

Planner Symmetry: Non-conducting of the shape.

* Flux will only be at the ends of surface; inside the surface, flux will be zero.

$\Phi_{\text{tot}} = \Phi_{\text{right}} + \Phi_{\text{left}}$ (Right = Left)

$\Phi_{\text{tot}} = 2EA \rightarrow 2EA = \frac{q}{\epsilon_0} \rightarrow 2EA = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}$

* Conducting sheet ki sirf ek side pr electric field exist krti jaise ki non-conducting ki har side pr krti.

③ Field at a point b/w two plates oppositely charged.

2 surfaces: Since two electric fields Planer Symmetry
 3 fields: $E = \frac{\sigma}{\epsilon_0}$ (final)



⇒ Field on any point on charged rod.

→ Cylindrical Symmetry: $\Rightarrow 22.7$ (Sider)

$q_{\text{enc}} = \epsilon_0 E \oint \cdot dA$
 $q_{\text{enc}} = \epsilon_0 E \cdot 2\pi r h$
 $\lambda = \frac{q}{L} = \epsilon_0 E \cdot 2\pi r$
 $E = \frac{\lambda}{2\pi \epsilon_0 r}$

* linear charge density (λ) = q/L
 * $A = 2\pi r h$ (cylinder)

⇒ Spherical Symmetry: From Sider. Conductor under electric field and charge is zero. Means; non-conductor ki andar bhi hote Imperial Notes

→ Deviation of Electric field due to dipole.

⇒ CHAP: 22 continue.

$E = E_{(+)} + E_{(-)}$
 $E = \frac{kq}{r_1^2} - \frac{kq}{r_2^2}$

$E = kq \left[\frac{1}{r_1^2} - \frac{1}{r_2^2} \right]$
 $r_1 = z - d/2$
 $r_2 = z + d/2$

$E = kq \left[\frac{1}{(z - d/2)^2} - \frac{1}{(z + d/2)^2} \right]$
 $E = kq \left[\frac{(z + d/2)^2 - (z - d/2)^2}{(z^2 - d^2/4)^2} \right]$
 $E = kq \left[\frac{z^2 + d^2/4 + zd - (z^2 - d^2/4 - zd)}{z^4 - (d/2)^4} \right]$
 $E = kq \left[\frac{2zd}{z^4 - (d/2)^4} \right]$

$E = \frac{q}{2\pi \epsilon_0} \left[\frac{2zd}{z^4 - (d/2)^4} \right]$
 $E = \frac{q}{2\pi \epsilon_0} \left[\frac{2zd}{z^4 - (d/2)^4} \right]$
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 $E = \frac{q}{2\pi \epsilon_0} \left[\frac{2zd}{z^4 - (d/2)^4} \right]$
 $E = \frac{q}{2\pi \epsilon_0} \left[\frac{2zd}{z^4 - (d/2)^4} \right]$

* $P = qd$ (dipole moment)

$E = \frac{1}{2\pi \epsilon_0} \frac{P}{z^3}$

electric field due to dipole at a point P.

→ Parallel Plate: $C = \frac{\epsilon_0 A}{d}$

Cylindrical Capacitance: $C = \frac{2\pi\epsilon_0 \cdot l}{\ln(b/a)}$

Spherical: $C = \frac{4\pi\epsilon_0 \cdot a \cdot b}{b-a}$

Day: ☆☆☆☆☆☆

Date: □□□□

MTWTFSS

Date: □□□□

Sample Problems: 1, 2, 3

Problems: 1, 2, 3, 4, 8, 9, 10, 11, 12, 13, 14, 15.

LECTURE-19

Thursday 7/11/24

⇒ CHAP: 25 "Capacitance"

charge $Q = CV$ → potential diff.

⇒ Parallel Plate Capacitance:

$C = \frac{Q}{V}$

we need PD.

$V = V_f - V_i = - \int_i^f E \cdot dr$

• path of integration is taken

$V = - \int_i^f E \cdot dr \cos \theta$

• Always against direction of field.

$V = - \int_i^f E \cdot dr \Rightarrow V = E \cdot r$

* Now for "Electric field" (E) we'll consider gaussian surface on only one plate.

$E = \frac{\sigma}{\epsilon_0}$

$\sigma = \frac{Q}{A}$

$E = \frac{Q}{\epsilon_0 A}$

• Put value of E in ①

$V = \frac{Qd}{\epsilon_0 A}$

• Now put value of 'V' in 'Q=CV'

$Q = C \left(\frac{Qd}{\epsilon_0 A} \right)$

$C = \frac{\epsilon_0 A}{d}$

Therefore capacitance depends upon: ① Area of plates, ② distance b/w plates, ③ Medium b/w plates (ϵ_0)

→ Cylindrical Capacitance:

$C = \frac{Q}{V}$

• From previous derivation:

$V = - \int E \cdot dr$

$V = - \int E \cdot dr$

* E of cylinder (from Gauss' law) = $\frac{\lambda}{2\pi\epsilon_0 r}$

$V = \int \frac{\lambda}{2\pi\epsilon_0 r} \cdot dr$

$$V = \frac{-\lambda}{2\pi\epsilon_0 b} \int_a^b \frac{1}{r} dr$$

$$V = \frac{-\lambda}{2\pi\epsilon_0 b} \left[\ln r \right]_a^b = \frac{-\lambda}{2\pi\epsilon_0 b} (\ln b - \ln a)$$

$$V = \frac{-\lambda}{2\pi\epsilon_0 b} \ln \left(\frac{b}{a} \right)$$

* Put 'V' into Q = CV

$$C = \frac{Q}{V} = \frac{\lambda \cdot \ln(b) - \ln(a)}{2\pi\epsilon_0 b \cdot \left(\frac{-\lambda}{2\pi\epsilon_0 b} \ln \left(\frac{b}{a} \right) \right)}$$

$$C = \frac{2\pi\epsilon_0 b}{\ln(b/a)}$$

$$C = \frac{2\pi\epsilon_0 b}{\ln(b/a)}$$

→ Spherical Capacitance:

$$C = \frac{Q}{V}$$

$$V = - \int_a^b E \cdot dr = - \int_a^b \frac{Q}{4\pi\epsilon_0 r^2} dr = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_a^b = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right)$$

$$V = \int_a^b E \cdot dr$$

$$V = \int_a^b E \cdot dr$$

* Electric field in sphere by assuming "Gaussian Surface" = $-\frac{kq}{r^2}$

$$V = - \int_a^b \frac{q}{4\pi\epsilon_0 r^2} dr$$

$$V = - \frac{q}{4\pi\epsilon_0} \int_a^b \frac{1}{r^2} dr$$

$$V = - \frac{q}{4\pi\epsilon_0} \left[-\frac{1}{r} \right]_a^b = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right)$$

$$V = - \frac{q}{4\pi\epsilon_0} \left[\frac{1}{b} - \frac{1}{a} \right]$$

* Reversing minus sign in formula 'kq/r^2' → (A)

* value of V from (A) into C = Q/V.

$$C = \frac{Q}{V} = \frac{q \left(\frac{1}{b} - \frac{1}{a} \right)}{\frac{q}{4\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right)} \rightarrow C = \frac{4\pi\epsilon_0 a \cdot b}{b - a}$$



→ Capacitance in Parallel:

$q_1 = C_1 V$, $q_2 = C_2 V$, $q_3 = C_3 V$, $q_n = C_n V$.

$q = q_1 + q_2 + q_3 = C_1 V + C_2 V + C_3 V$

$q = V(C_1 + C_2 + C_3)$

$I q = C_{eq} \cdot V \rightarrow \text{Equivalent (total)}$

$\therefore C_{eq} = C_1 + C_2 + C_3$

→ Capacitance in Series:

$V = \frac{q}{C}$

$V_1 = \frac{q}{C_1}$, $V_2 = \frac{q}{C_2}$, $V_3 = \frac{q}{C_3}$

$V = V_1 + V_2 + V_3 = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$

$V = q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$

$V = \frac{q}{C_{eq}}$

$\therefore \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

Sample Problem: 1, 2

Problems: 1, 2, 5, 7, 8, 10, 11, 12, 13, 14, 15, 17, 18, 23, 29, 31, 32.

→ CHAP # 26: "Current & Resistance".

- Electric current: Flow of e^- (charges) from $+$ to $-$ ve
- Conventional current: Flow of e^- (charges) from $+$ to $-$ ve

* Since charge is conserved, so current also conserved.

$I = \frac{Q}{t}$

→ Current Density: (J) moving $+$ ve charges makes J point in the same direction

$J = \frac{I}{A}$

→ Area of surface.

(will be '2πr' in case of wire)

Total Current in this shape:

$I_{\text{tot}} = \int J \cdot dA$

→ Drift Speed: (V_d)

$$I = I/A$$

$$I = q/t \rightarrow (A)$$

$$q = neV \rightarrow [q = neAt]$$

Area and length of conductor.
Volume of conductor.

Since $s = vt$ so, $s = v_0 t$ (drift velocity).
 $s = L$ (length of conductor); $v = v_0$ (drift velocity).

$$L = v_0 t \rightarrow [t = \frac{L}{v_0}]$$

Putting value of 't' and 'q' in eq (A)

$$I = \frac{qeAt}{L} \rightarrow [I = qeA \cdot (V_d)]$$

Putting value of I in $J = I/A$

$$J = \frac{qeA V_d}{A} \rightarrow [J = qe V_d]$$

$$\therefore V_d = \frac{J}{qe}$$

* value of V_d is different at each point.
No. of e^- per unit volume.

Resistance: Object's Property.

Resistivity: Material of object.

$$\rho = \frac{E}{J}$$

Conductivity: Flow of charges. (σ)

$$\sigma = \frac{1}{\rho} = \frac{J}{E}$$

→ Formula for Resistance (to see dependence upon factors)

$$\rho = E/J \rightarrow (A)$$

$V = E \cdot l$ (from capacitance chapter).

Put 'E' into (A).

Put $J = I/A$ into (A)

$$\rho = \frac{V/l}{I/A}$$

$$\therefore V/I = R$$

$$\rho = \frac{RA}{L} \rightarrow [R = \frac{\rho L}{A}]$$

$\therefore$ Resistance depends upon material, length, area of object.

LECTURE-21

Thursday

21/11/24

⇒ CHAP # 28 "Magnetic Field".

$E = \frac{V}{d}, E = (N/C) B \rightarrow V = \frac{V}{q} \rightarrow eV = \frac{1}{2}mv^2$
 $v = \sqrt{\frac{2eV}{m}}$

Problems: 1, 3, 4, 5, 13, 14, 21, 23, 24, 25, 39, 40, 41, 42, 45.

- ★ Negative charge: Flemings Left hand Rule
- ★ Positive: Right hand rule.

⇒ Grating charged Particle.

★ $F_B = F_c$

$t = \frac{s}{v}$
 $qvB = \frac{mv^2}{r}$

For circle: $s = 2\pi r$

$t = \frac{2\pi r}{v}$

★ Put 'v' from (A)

$t = \frac{2\pi \cdot mv}{qvB} \rightarrow t = \frac{2\pi m}{qB}$

$\rightarrow f = 1/t$

$\omega = 2\pi f$

$\Delta B = \Delta A$

$\Delta A = \Delta B$

$A = IV$

LECTURE-22

Thursday

26/11/24

⇒ CHAP # 29 "Mag. field due to current".

Mag. field $\leftarrow B = \frac{\mu_0 I}{2\pi R}$

for straight wire $\rightarrow$ distance from center to specific point.

Anti-clockwise + Out of page

⇒ Ampere's law: = the (vice versa).

$\oint B \cdot ds = \mu_0 I_{enc}$
 $I_{enc} = I_1 + I_2$
 $I_{enc} = NI$

★ Use right hand thumb rule to curl fingers in direction of B gives thumb as the direction of current. Since I_1 is in direction of thumb (i.e. outwards); so I_1 is true $I_2 = true$

⇒ Derivation of $B = \mu_0 I / 2\pi R$. (Biot Savart)

$\oint B \cdot ds = \mu_0 I_{enc}$

★ since the arcs and mag. field line direction is same, so $\theta = 0, \cos \theta = 1$.

$\oint B \cdot ds \cos \theta = \mu_0 I_{enc}$
 $\star ds = 2\pi R$ (circumference)

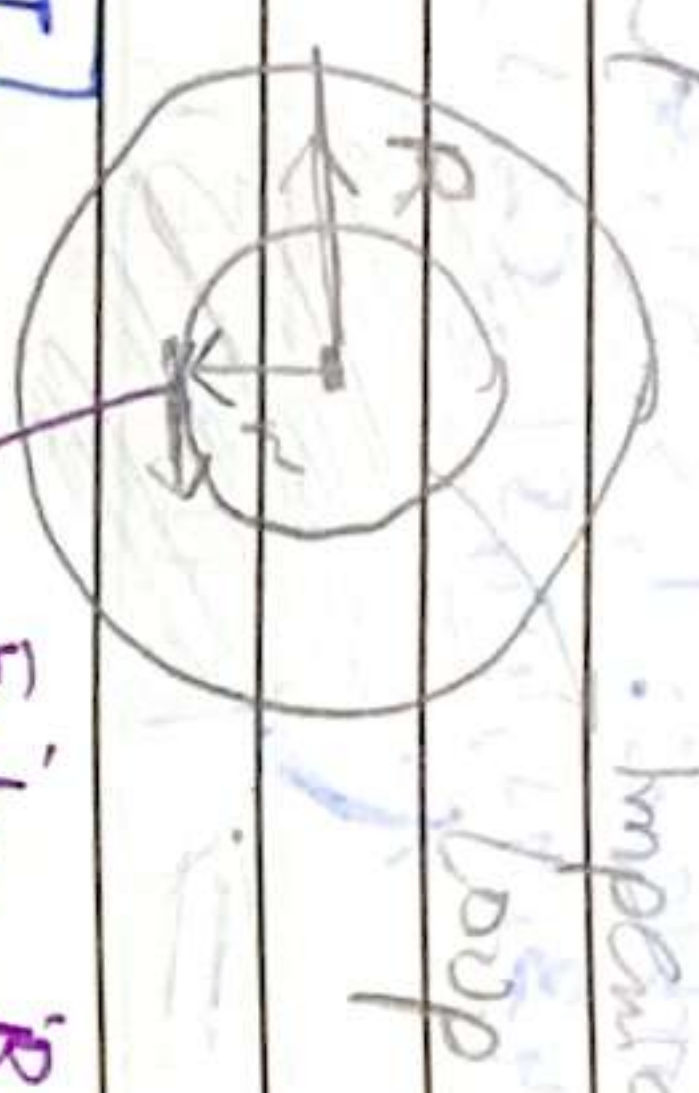
$B (2\pi R) = \mu_0 I_{enc}$

⇒ Very field inside straight wire carrying current

$B \cdot ds = \mu_0 I_{enc}$

$B \cdot \phi \cdot ds = B(2\pi r)$

$I_{enc} = \left(\frac{I r^2}{R^2} \right) I \rightarrow I_{enc} = \frac{r^2}{R^2} I$



* value of I_{enc} in ampere's law.
* since we using $2\pi r$ for entire surface so $\int$ sign vanishes.

$B(2\pi r) = \mu_0 \left(\frac{r^2}{R^2} \right) I$

$B = \left(\frac{\mu_0 r}{2\pi R^2} \right) I$

Force b/w 2 parallel current carrying wires

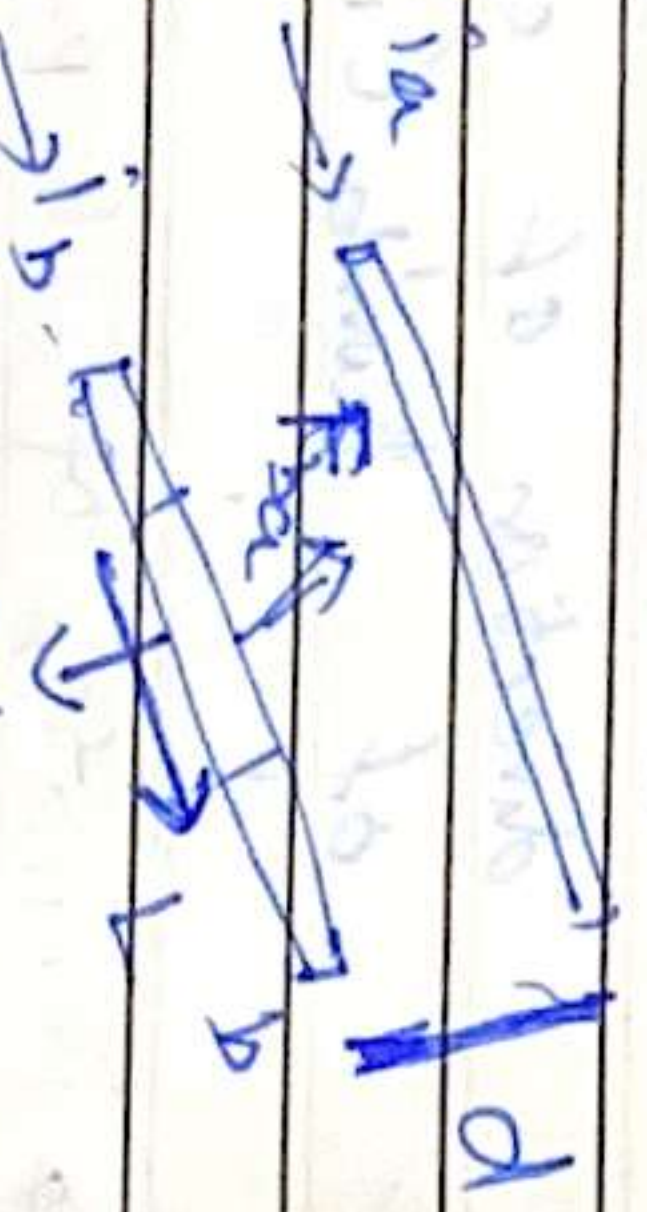
* Suppose force on wire 'b' by wire 'a'

$F_{ba} = \frac{\mu_0}{4\pi} \frac{I_a I_b L}{R_a} \sin \theta = 90^\circ$

$F_{ba} = I_b L B_a$

$B_a = \frac{\mu_0 I_a}{2\pi d}$

$F_{ba} = \frac{\mu_0 I_a I_b L}{2\pi d}$



Solenoid

by using Ampere's Law

$\oint B \cdot ds = \mu_0 I_{enc}$

1. Loop: abcd a

$\oint B \cdot ds = \int_a^b B \cdot ds + \int_b^c B \cdot ds + \int_c^d B \cdot ds + \int_d^a B \cdot ds$

$\cos \theta = 1$

$\int B \cdot ds = B h$

$B h = \mu_0 I_{enc}$

$I_{enc} = N \cdot I \rightarrow I_{enc} = n h I$

$B h = \mu_0 I n h$

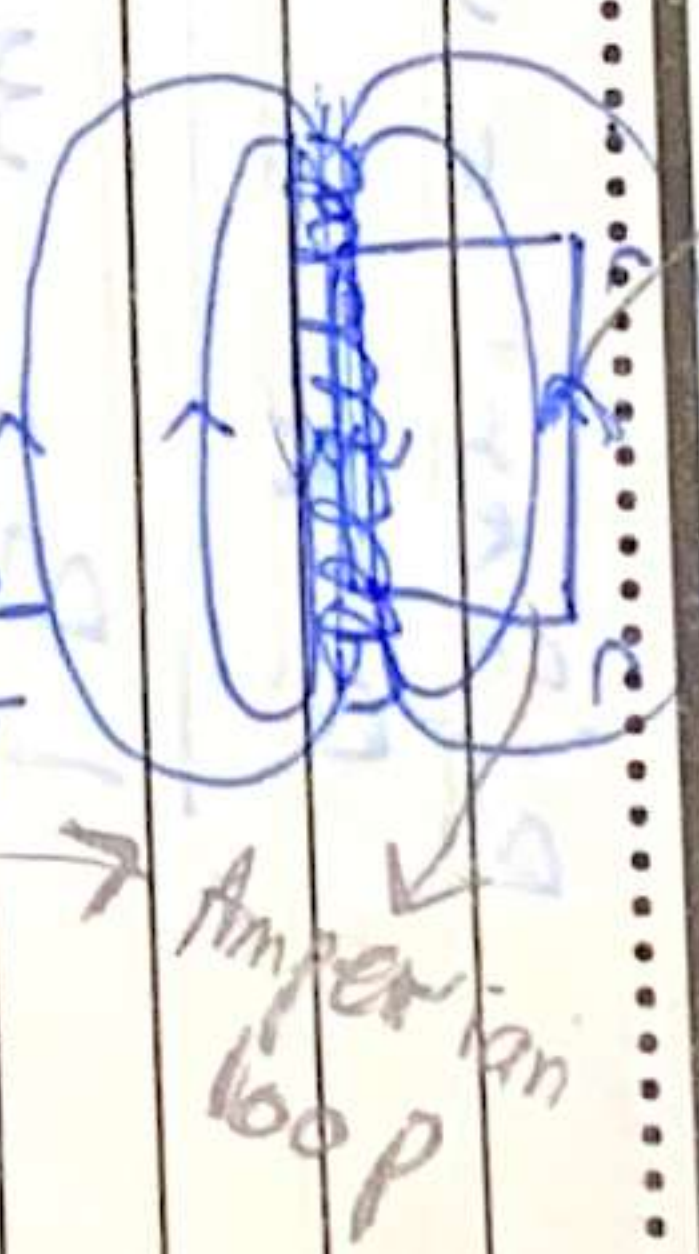
$B = \mu_0 I n$

Toroid

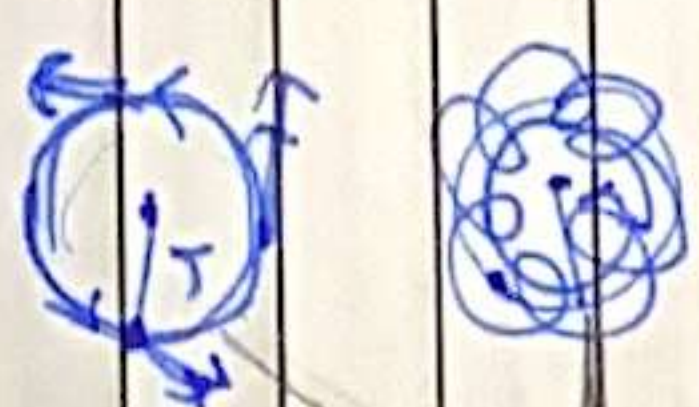
By using Ampere's Law

$\oint B \cdot ds = \mu_0 I_{enc}$

$B \int ds = \mu_0 I_{enc}$



approx. zero field since far from solenoid.



Amperian loop

